

# Zeros of weighted partial sums of completely multiplicative functions

## Dirichlet characters, the Euler factor at 2 and Conrey's sine series

Carles Marín\*

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### Abstract

A completely multiplicative function  $\varepsilon$  with values  $\pm 1$  can make  $\sum_{n \leq m} n \varepsilon(n)$  vanish only when  $m \equiv 0, 3 \pmod{4}$ . We show that some  $\varepsilon$  does for every such  $m$ . The primes of  $(m/2, m]$  enter the sum only as themselves: a second moment fixes the other signs with a small remainder, and the signed sums of those primes cover it, by interval certificates up to  $10^8$  and by Sárközy's progressions in subset sums beyond. Every finite prefix of signs extends to a zero at all large admissible  $m$ , so the functions with infinitely many zeros form a dense  $G_\delta$  of probability zero. Quadratic characters, extended completely multiplicatively, give explicit points of it: a digit recursion propagates their zeros in base  $q$ , for  $q \equiv 1 \pmod{4}$ , for  $q \equiv 7 \pmod{8}$  — where the half sum of Jacobi and Dirichlet, with its Euler factor at 2, does the work — and along loops of the recursion, as for class number one. For odd characters, the digits that propagate zeros are the intervals on which Conrey's sine series of the Legendre symbol, proposed as a route to the Riemann hypothesis, vanishes identically. The same Euler factor decides which components of the moment matrix of the ring of character values vanish on its diagonal, which is where this note comes from.

**Keywords:** completely multiplicative functions; weighted sums; exact zeros; signed sums of primes; subset sums; random multiplicative functions; Dirichlet characters; Legendre symbol; digit recursion; class number; Euler factor; Riemann hypothesis.

## 1 Half sums since 1832, and a question of signs

In 1832 Jacobi stated, and in 1838 Dirichlet proved, that for a prime  $p \equiv 3 \pmod{4}$ ,  $p > 3$ , the class number of  $\mathbb{Q}(\sqrt{-p})$  is a half sum of Legendre symbols divided by an Euler factor at 2,

$$h(-p) = \frac{1}{2 - \left(\frac{2}{p}\right)} \sum_{0 < n < p/2} \left(\frac{n}{p}\right)$$

[12], [13, Werke I, p. 365]. The factor  $2 - (2 | p)$  is the whole difference between the half sum and the full one, and it will reappear. Weighted half sums came later. Maillet asked in 1913 whether a determinant of residues modulo  $p$  can vanish [14]; Carlitz and Olson evaluated it in 1955 as  $\pm p^{(p-3)/2} h^-$ , remarking that Chowla and Weil had had the result before them [15]; Jager and Lenstra described the linear relations among the cosecant values behind it [30], Girstmair their annihilators for every odd conductor [31], and Kučera and Kanemitsu–Kuzumaki wrote relative class numbers as determinants of that kind [34, 33]. In that lane the question is when a half sum vanishes, and the answer is again the prime 2.

A different question is asked of the same sums once the character is replaced by an arbitrary completely multiplicative sign. Pólya observed in 1919 that the summatory function of the Liouville function seemed never to be positive [16]; Turán showed in 1948 that positivity of its logarithmic companion would imply the Riemann hypothesis [17]; Haselgrove disproved both in 1958 [18, 19]. Erdős asked, around 1932 and in print in 1957, whether every  $\pm 1$  sequence has unbounded sums along homogeneous progressions [20, Problem 9], and Tao proved that it

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\*Independent researcher, [karlesmarin@gmail.com](mailto:karlesmarin@gmail.com). Written with Claude (Anthropic) as an assistant.

does [21]; Borwein, Choi and Coons described the mean values and partial sums of completely multiplicative signs [22], and Klurman, Munsch and Sun found completely multiplicative functions whose signed harmonic sums come exponentially close to zero along a sequence [23]. And Conrey showed in 2024 that the positivity near 0 of the sine series  $\sum \lambda(n) \sin(2\pi nx)/n^2$  would imply the Riemann hypothesis, proposing to reach it through Legendre symbols, which agree with  $\lambda$  on long initial segments [4]. These are questions of size and of sign. An *exact* zero of a *weighted* sum, at *every* admissible length, is not among them.

The tool that decides it comes from a third lane. Sárközy showed that the subset sums of a dense set of integers contain a long arithmetic progression [24], Lev sharpened the constants [25], and Bringmann and Wellnitz stated the result in the explicit form used below [26, Thm. 4.23]; Szemerédi and Vu, and Conlon, Fox and Pham, pushed the thresholds [27, 28].

This note stands where the three meet, as Figure 1 draws it. Its main result, Theorem 1, says that every admissible length carries an exact zero, and it is decided with the third lane. The question came from the first: the components of the moment matrix of [29] are weighted half sums of characters, which on the diagonal vanish by the Euler factor of the first lane and off it, for a quadratic character, are the weighted sums of the second (Section 4). Along a single function the lanes meet once more: many quadratic characters are among the rare functions with infinitely many zeros, and for  $q \equiv 7 \pmod{8}$  the first lane's Euler factor is what puts them there (Corollary 8); and the digits that carry those zeros are the intervals on which Conrey's sine series of the Legendre symbol vanishes identically (§3.2).

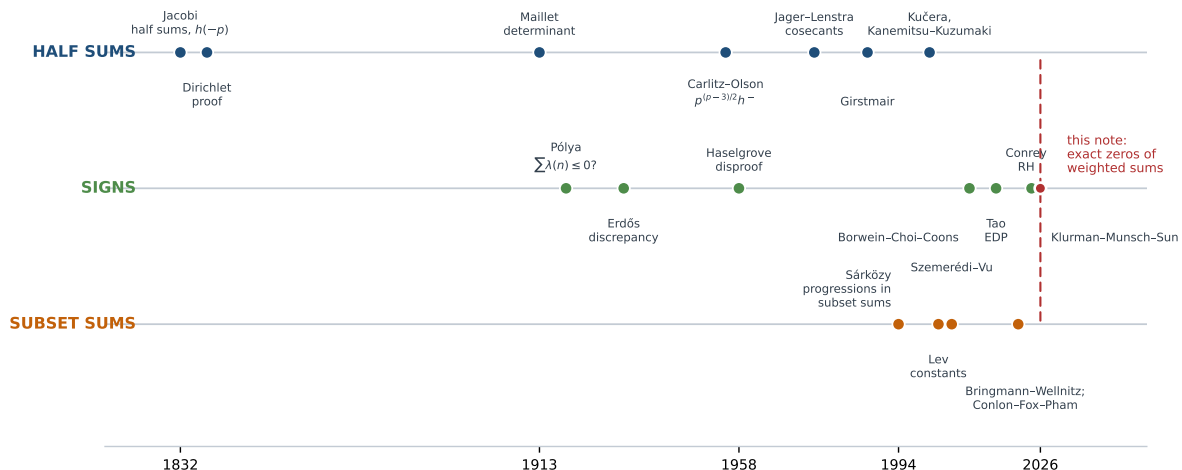


Figure 1: Three lanes, and this note at the crossing. *Half sums*: from Jacobi's class number formula to the determinants of Mailliet, Carlitz–Olson and Kučera; the question is when they vanish, and the answer is the prime 2. *Signs*: the sizes and signs of sums of completely multiplicative  $\pm 1$  functions, from Pólya's observation to Tao's proof of Erdős's discrepancy problem and Conrey's route to the Riemann hypothesis through Legendre symbols. *Subset sums*: arithmetic progressions among subset sums, the tool that turns a size estimate into an exact zero.

*Which weighted sums vanish, at which lengths, and along which functions?*

The table answers, one row per question. As in [29], a statement is either proved or marked *measured* with the number of cases behind it, and what is left open is marked  $\triangleright$  and collected at the end.

| question                                                             | answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| How does a zero appear at every admissible length? (§2)              | the primes of $(m/2, m]$ enter once: a second moment fixes the rest, and their signed sums cover the remainder — by certificates up to $10^8$ , by Sárközy's progressions beyond. Theorem 1, Table 1, Figure 2                                                                                                                                                                                                                                                                 |
| How often does one function vanish? (§3)                             | infinitely often on a dense $G_\delta$ , but with probability zero: the weight makes the sum jump over 0. Proposition 4, Theorem 5                                                                                                                                                                                                                                                                                                                                             |
| Which functions do? (§3)                                             | quadratic characters, through a digit recursion: for $q \equiv 1 \pmod{4}$ , for $q \equiv 7 \pmod{8}$ by the Euler factor at 2 again, for class number one, wherever the recursion closes a loop, and for $\chi(q) = -1$ as soon as there is one zero; for odd characters of any order, at $(q^j - 1)/2$ , exactly when $\chi(2) = 1$ . The null cells of Conrey's sine series are the same digits. Theorem 6, Corollaries 8 and 10, Propositions 9 and 11, Table 2, Figure 3 |
| Where does the question come from? (§4)                              | from the moment matrix of the ring of character values in [29], whose components are weighted half sums of characters. The rest of this table is about them.                                                                                                                                                                                                                                                                                                                   |
| Which components of the moment matrix vanish on the diagonal? (§4.2) | those at the odd characters with $\chi(2) = 1$ , by an Euler factor at 2: $d/t$ of them for a prime conductor, and $d/t$ is the rank defect for every odd conductor. Theorem 7, Corollary 13, Theorem 14                                                                                                                                                                                                                                                                       |
| Which rows off the diagonal can be occupied? (§4.3)                  | exactly those with $\left(\frac{2}{n}\right) = 1$ , that is $m \equiv 0, 3 \pmod{4}$ : parity on one side, Theorem 1 on the other. Proposition 16, Corollary 17                                                                                                                                                                                                                                                                                                                |

## 2 Every admissible length

Here  $\varepsilon$  is a completely multiplicative function  $\mathbb{N} \rightarrow \{\pm 1\}$ , determined by its values at the primes, and

$$S_\varepsilon(m) = \sum_{n \leq m} n \varepsilon(n).$$

Changing one sign changes  $S_\varepsilon(m)$  by an even amount, so  $S_\varepsilon(m) \equiv m(m+1)/2 \pmod{2}$ , and a zero is possible only when  $m \equiv 0$  or  $3 \pmod{4}$ ; call these lengths *admissible*. Every admissible length carries a zero.

**Theorem 1** (every admissible length). *For every  $m \geq 3$  with  $m \equiv 0$  or  $3 \pmod{4}$  there is a completely multiplicative  $\varepsilon : \mathbb{N} \rightarrow \{\pm 1\}$  with*

$$\sum_{n \leq m} n \varepsilon(n) = 0.$$

### 2.1 The large primes enter once

Everything rests on one observation. A prime  $q \in (m/2, m]$  divides no  $n \leq m$  other than  $q$  itself, so its sign enters  $S_\varepsilon(m)$  once, with weight  $q$ . For a set  $Q$  of such primes,

$$S_\varepsilon(m) = R_Q + \sum_{q \in Q} q \varepsilon(q), \tag{1}$$

where  $R_Q$  depends only on the signs outside  $Q$ . Since  $S_\varepsilon(m)$  is even at admissible lengths,  $R_Q \equiv P_Q \pmod{2}$ , with  $P_Q = \sum_{q \in Q} q$ . The signed sums  $\sum \pm q$  are the numbers  $P_Q - 2x$ ,  $x$  a subset sum of  $Q$ . So the proof has two halves: make  $|R_Q|$  small by the signs outside  $Q$ , and

show that the subset sums of  $Q$  contain a long interval  $[\alpha, P_Q - \alpha]$ ; then every integer of the parity of  $P_Q$  in  $[-D, D]$ ,  $D = P_Q - 2\alpha$ , is a signed sum, and  $|R_Q| \leq D$  gives a zero.

**Lemma 2** (the second moment). *Give the primes outside  $Q$  independent uniform signs. Then*

$$\mathbb{E} R_Q^2 \leq \frac{m^3(\log m + 13)}{9},$$

so some choice has  $|R_Q| \leq B_*(m) := \frac{1}{3}m^{3/2}\sqrt{\log m + 13}$ .

*Proof.*  $\mathbb{E}\varepsilon(n)\varepsilon(k)$  is 1 if  $nk$  is a square and 0 otherwise. Writing  $n = da^2$  with  $d$  squarefree,  $\mathbb{E} S_\varepsilon(m)^2 = \sum_d b_m(d)^2$  with  $b_m(d) = d \sum_{j \leq \sqrt{m/d}} j^2$ , and  $\mathbb{E} R_Q^2$  is the same sum without the terms  $d = q \in Q$ , whose class contains only  $n = q$ . With  $J = \lfloor \sqrt{m/d} \rfloor$ ,  $\sum_{j \leq J} j^2 \leq J^3/3 + J^2/2 + J/6$ , and  $dJ^2 \leq m$ ,  $dJ \leq \sqrt{md} \leq m$  give  $b_m(d) \leq m^{3/2}/(3\sqrt{d}) + 2m/3$ . Squaring and summing over  $d \leq m$  with  $\sum 1/d \leq 1 + \log m$ ,  $\sum d^{-1/2} \leq 2\sqrt{m}$  and at most  $m$  terms gives  $m^3((1 + \log m) + 8 + 4)/9$ .  $\square$

**Lemma 3** (intervals of subset sums). *Write  $\Sigma(A)$  for the set of subset sums of a finite set  $A$  of positive integers.*

- (a) *If  $\Sigma(A) \supseteq [\alpha, \beta]$  and  $x \notin A$ ,  $0 < x \leq \beta - \alpha + 1$ , then  $\Sigma(A \cup \{x\}) \supseteq [\alpha, \beta + x]$ . As  $\Sigma(A)$  is symmetric under  $x \mapsto P_A - x$ ,  $\Sigma(A) \supseteq [\alpha, \beta]$  with  $\beta \geq P_A/2$  implies  $\Sigma(A) \supseteq [\alpha, P_A - \alpha]$ .*
- (b) (pairs) *Let  $q_0, u_1, v_1, \dots, u_t, v_t$  be distinct odd integers,  $d_j = (v_j - u_j)/2$  with  $1 \leq d_j \leq 1 + d_1 + \dots + d_{j-1}$ ,  $H = \sum d_j$ ,  $U = \sum u_j$ , and  $q_0 \leq 2H + 1$ . Then  $\Sigma(\{q_0, u_j, v_j\}) \supseteq [U + q_0 - 1, U + 2H + 1]$ .*
- (c) (Sárközy, with Lev's constants [26, Thm. 4.23]) *If  $X$  is a set of  $n$  positive integers with maximum  $x^*$  and  $4x^* \leq \ell \leq n^2/(12 \log(4x^*/n))$ , then  $\Sigma(X)$  contains an arithmetic progression of  $\ell$  terms and step  $s \leq 4x^*/n$ , each term a sum of at most  $6\ell/n$  distinct elements of  $X$ .*

*Proof.* (a) is immediate. (b) The condition on the  $d_j$  makes their subset sums exactly  $\{0, \dots, H\}$ ; choosing one element of each pair gives  $U + 2\{0, \dots, H\}$ , and adding  $q_0$  or not fills the other parity: an integer  $x$  of the interval with  $x - U$  even is  $U + 2k$ ,  $k \leq H$ , and one with  $x - U$  odd is  $U + q_0 + 2k$  with  $0 \leq k \leq H$ . (c) is the cited theorem.  $\square$

## 2.2 The proof

*Proof of Theorem 1.* Table 1 summarises the three ranges.

*Up to 20000.* For each admissible  $m \leq 20000$  a function  $\varepsilon$  with  $S_\varepsilon(m) = 0$  is exhibited and the sum is recomputed exactly: for all but four lengths by starting from  $\varepsilon(2) = \varepsilon(3) = -1$  (with at most two signs at primes  $\leq 7$  changed), balancing the primes of  $(\sqrt{m}, m/2]$  and solving (1) exactly with the primes of  $(m/2, m]$ ; the lengths 12, 15, 19, 23 by exhaustive search.

*From 5000 to  $10^8$ .* Cut  $[5000, 10^8)$  into 104 blocks  $[a_j, b_j]$ ,

$$a_1 = 5000, \quad b_j = \min(\lfloor 11a_j/10 \rfloor, 10^8 - 1), \quad a_{j+1} = b_j + 1,$$

and let  $Q$  be the primes of  $(b_j/2, a_j]$ , which lie in  $(m/2, m]$  for every  $m$  in the block; the other primes of  $(m/2, m]$  are among the random signs of Lemma 2. Each block carries a certificate: a prime  $q_0 \in Q$  and disjoint pairs in  $Q$  as in Lemma 3(b), with  $2H - q_0 + 2 \geq b$ . Every other prime of  $Q$  is at most  $a \leq b$ , so Lemma 3(a) extends the interval to  $[\alpha, P_Q - \alpha]$  with  $\alpha = U + q_0 - 1$ . The block is proved for all its  $m$  once  $D = P_Q - 2\alpha$  exceeds  $B_*(b) \geq B_*(m)$ , which is the integer inequality  $90D^2 > b^3(130 + 7e)$  with  $e = \lceil \log_2 b \rceil$  (as  $\log b < 7e/10$ ). All 104 certificates pass; the bound  $b^3(130 + 7e)/(90D^2)$  on  $B_*(b)^2/D^2$  is at most 0.445, reached at the first block.

From  $10^8$  on. Let  $M \geq 10^8$ ,  $L = \log M$ ,  $Q$  the  $N$  primes of  $(M/2, M]$ . By Dusart's bounds  $x/(\log x - 1) \leq \pi(x)$  for  $x \geq 5393$  and  $\pi(x) \leq x/(\log x - 1.1)$  for  $x \geq 60184$  [5, Thm. 6.9, (6.6)],  $N \geq M/(2L)$  and  $P_Q \geq NM/2 \geq M^2/(4L)$ . Let  $X \subset Q$  have  $n = \lfloor N/8 \rfloor \geq M/(17L)$  elements and take  $\ell = 4M$ : the hypothesis of Lemma 3(c) follows from  $n^2 \geq 48M \log(4M/n)$ , which holds once  $M \geq 13872L^2 \log(68L)$ , that is from  $M \approx 2.9 \cdot 10^7$  (from  $4.4 \cdot 10^7$  if the logarithm of Lemma 3(c) is read to base 2). This gives a progression of step  $s \leq 4M/n$  whose terms are at most  $24M^2/n$ . Add  $s - 1$  further primes of  $Q$ : they are units modulo  $s$ , so their subset sums meet every class, and the progression becomes an interval  $[\alpha, \alpha + w]$  with  $\alpha \leq 28M^2/n$  and  $w \geq (\ell - 1)s - (s - 1)M \geq M$ . The remaining primes, each at most  $M$ , extend it by Lemma 3(a); they are  $N - n - s + 1 \geq \frac{7}{8}N - s$  primes above  $M/2$ , while  $X$  and the  $s - 1$  added primes weigh at most  $(\frac{1}{8}N + s)M$ , so the remaining ones carry more than half of  $P_Q$  and the symmetry closes the interval to  $[\alpha, P_Q - \alpha]$ . With  $K = 2\alpha \leq 56M^2/n$  and Lemma 2,

$$\frac{B_*(M) + K}{P_Q} \leq \frac{4}{3} \frac{L\sqrt{L+13}}{\sqrt{M}} + \frac{3808 L^2}{M} < 0.027 \quad (M \geq 10^8),$$

both terms decreasing in  $M$ . So  $|R_Q| < P_Q - K$  and a zero exists.  $\square$

| range of $m$          | method                                  | certificate            | margin                  |
|-----------------------|-----------------------------------------|------------------------|-------------------------|
| $3 \leq m \leq 20000$ | explicit $\varepsilon$ , sum recomputed | 10000 lengths          | exact                   |
| $5000 \leq m < 10^8$  | Lemmas 2, 3(a,b)                        | 104 blocks, 2128 pairs | $B_*^2/D^2 \leq 0.445$  |
| $m \geq 10^8$         | Lemmas 2, 3(c), Dusart                  | none                   | $(B_* + K)/P_Q < 0.027$ |

Table 1: The proof of Theorem 1, range by range. The ranges overlap on purpose. The certificates are generated and checked, in integer arithmetic, by a script that runs in seconds.

Figure 2 shows the margin of each of the three ranges. The seam at  $10^8$  is not an accident of taste. The pairs of Lemma 3(b) must start with  $d_1 = 1$ , a pair of twin primes inside  $(b/2, a]$ : every block has one, but no theorem says every interval will, so the certificates cannot be turned into a proof for all  $m$ . Sárközy's progression needs no such luck, and it is available from about  $2.9 \cdot 10^7$ ; the certificates overlap it with room.

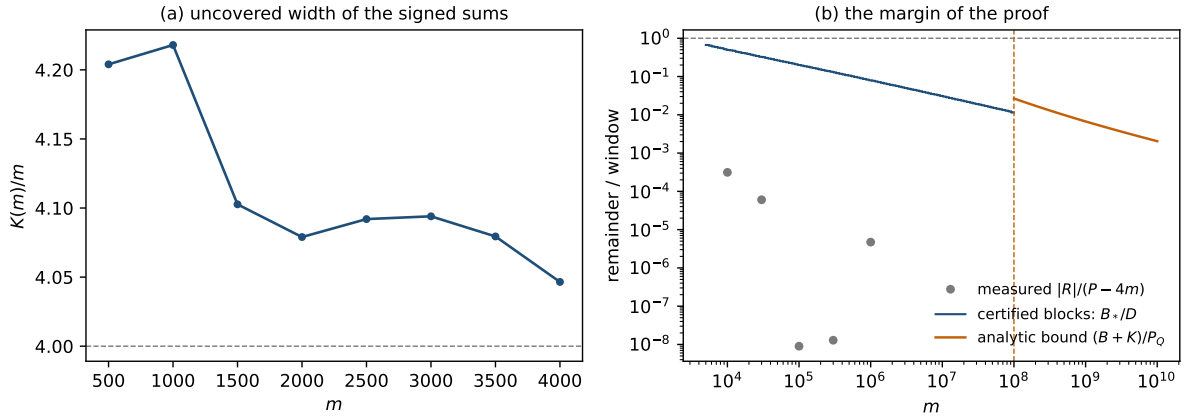


Figure 2: Why every admissible length closes. (a) The width  $K(m)$  that the signed sums of the primes of  $(m/2, m]$  leave uncovered at each end, measured exactly: about  $4m$ . (b) The margin of the proof: the remainder over the window, measured to  $10^6$  (grey), the certified bound  $B_*/D$  in each of the 104 blocks of  $[5000, 10^8]$  (blue), and the analytic bound from  $10^8$  on (orange). Every curve stays below 1.

*Measured:* Lemma 2 exactly,  $\mathbb{E} S_\varepsilon(m)^2$  computed from the squarefree kernels at 24176 values of  $m \leq 10^8$ , all block ends included: the ratio to the bound never exceeds 0.43. Lemma 3(b)

against the true subset sums of the first eight blocks: the uncovered end is 3.6 to 3.9 times smaller than the certified  $\alpha$ .

### 3 Along one function

Theorem 1 lets  $\varepsilon$  depend on  $m$ . Fix it instead, and ask how often  $S_\varepsilon(m)$  vanishes.

**Proposition 4** (prefixes). *Let  $F$  be a finite set of primes and  $\varepsilon_0 : F \rightarrow \{\pm 1\}$ . There is  $M(F)$  such that for every admissible  $m \geq M(F)$  some  $\varepsilon$  with  $\varepsilon|_F = \varepsilon_0$  has  $S_\varepsilon(m) = 0$ . For  $F = \{2, 3, 5, 7\}$  one may take  $M(F) = 60000$ .*

*Proof.* For  $m > 2 \max F$  the set  $F$  is disjoint from  $Q$ . Conditioning on the event  $\varepsilon|_F = \varepsilon_0$ , of probability  $2^{-|F|}$ , multiplies  $\mathbb{E} R_Q^2$  by at most  $2^{|F|}$ , since  $R_Q^2 \geq 0$ ; the proof of Theorem 1 then goes through with  $2^{|F|/2} B_*$  in place of  $B_*$ , and both  $B_*/P_Q$  and  $K/P_Q$  tend to 0. For  $F = \{2, 3, 5, 7\}$  the factor is 4: every block with  $b \geq 60000$  satisfies  $90D^2 > 16b^3(130 + 7e)$ , and beyond  $10^8$ ,  $4B_*/P_Q + K/P_Q < 0.07$ .  $\square$

**Theorem 5** (a generic set of probability zero). *On  $\Omega = \{\pm 1\}^{\mathcal{P}}$ , the sign patterns on the primes, let  $\mathcal{Z}_\infty$  be the set of  $\varepsilon$  with  $S_\varepsilon(m) = 0$  for infinitely many  $m$ . With the product topology,  $\mathcal{Z}_\infty$  is a dense  $G_\delta$ . With independent fair signs,  $\Pr(\mathcal{Z}_\infty) = 0$ .*

*Proof.* The set  $U_M$  of  $\varepsilon$  with a zero at some admissible  $m \geq M$  is open, each condition involving finitely many signs, and dense by Proposition 4;  $\mathcal{Z}_\infty = \bigcap_M U_M$ . For the measure, condition on every sign outside the primes  $Q$  of  $(m/2, m]$ : by (1),  $S_\varepsilon(m) = R_Q + \sum_{q \in Q} q \varepsilon(q)$  with distinct weights, and the theorem of Sárközy and Szemerédi [10], [9, Thm. 1.1] bounds the probability of any single value by  $O(|Q|^{-3/2}) = O((\log m/m)^{3/2})$ . The series converges, and Borel–Cantelli gives  $\Pr(\mathcal{Z}_\infty) = 0$ .  $\square$

The weight is what makes this true. Without it the steps of  $\sum_{n \leq m} \varepsilon(n)$  are  $\pm 1$ , a change of sign forces a zero, and almost surely there are infinitely many [3]; with the logarithmic weight  $1/n$  the question becomes one of small values [23]. With the weight  $n$  the sum jumps over 0.

#### 3.1 Characters in the generic set

Theorem 5 says that a function with infinitely many zeros is exceptional in measure. Many natural functions are among the exceptions. For an odd prime  $q$  and  $s = \pm 1$ , let  $\chi = \chi_{q,s}$  be the completely multiplicative function equal to the Legendre symbol  $(\cdot | q)$  on the integers prime to  $q$ , with  $\chi(q) = s$ . Put

$$C(r) = \sum_{r' \leq r} \chi(r'), \quad T(r) = \sum_{r' \leq r} r' \chi(r') \quad (0 \leq r < q), \quad T_q = T(q-1).$$

**Theorem 6** (a digit recursion). *For  $k \geq 0$  and  $0 \leq r < q$ ,*

$$S_\chi(qk + r) = s q S_\chi(k) + k(T_q + q C(r)) + T(r).$$

*Proof.* The  $n \leq qk + r$  prime to  $q$  are  $jq + r'$  with  $j < k$ ,  $1 \leq r' < q$ , and  $kq + r'$  with  $r' \leq r$ ; as  $\chi(jq + r') = \chi(r')$  and  $C(q-1) = 0$ , they contribute  $kT_q + kq C(r) + T(r)$ . The multiples  $jq$ ,  $j \leq k$ , contribute  $\sum_j jq s \chi(j) = sq S_\chi(k)$ .  $\square$

Theorem 6 and the propagation of zeros in Corollary 8 below are formalized in Lean 4 with Mathlib for every completely multiplicative  $\chi : \mathbb{N} \rightarrow \mathbb{Z}$  that is periodic modulo  $q$  on the residues prime to  $q$  and sums to 0 over a period, together with Lemma 3(b), which needs only  $q_0$  odd; the

file travels with this note and depends on no axiom beyond the standard three. The unweighted analogue is classical: Borwein, Choi and Coons read the partial sums of  $\chi_{q,1}$  off the base- $q$  digits [22, Thm. 9], and Klurman, Mangerel, Pohoata and Teräväinen give the recursion for any value at  $q$  [8, Lemma 2.8]. The unweighted partial sums of Legendre symbols are also studied as random-like paths [6] and for their sign at fixed fractions of  $q$  [7]. The weight adds the term  $k(T_q + qC(r))$ , and with it the zeros.

One ingredient is classical in form. For an odd character the weighted half sum has a closed expression through  $B_{1,\chi}$  and the Euler factor at 2; in the form of a sum against the shifted sawtooth it is in Kanemitsu and Kuzumaki [33, proof of Thm. 1] and in Kučera [34, Ex. 2]. We state it for the half sum, with a short proof, because the vanishing criterion is what the zeros need, and it is also the diagonal of §4.2.

**Theorem 7** (the weighted half sum). *Let  $f$  be odd and let  $\chi$  be an odd Dirichlet character modulo  $f$ . Then*

$$\sum_{c=1}^{(f-1)/2} c \chi(c) = \frac{f}{2} B_{1,\chi} (\chi(2)^{-1} - 1).$$

*If moreover  $\chi$  is primitive, so that  $B_{1,\chi} \neq 0$ , the sum vanishes if and only if  $\chi(2) = 1$ .*

*Proof.* Write  $T = \sum_{c < f/2} c \chi(c)$ ,  $U = \sum_{c < f/2} \chi(c)$ ,  $W = \sum_{c \text{ even}} \chi(c)$  and  $X = \sum_{c \text{ even}} c \chi(c)$ , all over  $1 \leq c < f$ . Pairing  $b \mapsto f - b$  exchanges odd and even residues, since  $f$  is odd, and  $\chi(f-b) = -\chi(b)$ ; hence  $\sum_{b \text{ odd}} b \chi(b) = -fW + X$ , and adding  $X$  gives  $fB_{1,\chi} = 2X - fW$ , that is  $X = \frac{f}{2}(B_{1,\chi} + W)$ . Substituting  $c = 2c'$  in the even sums,  $W = \chi(2)U$  and  $X = \chi(2) \cdot 2T$ . Finally  $\sum_b \chi(b) = 0$  splits as  $\sum_{\text{odd}} + \sum_{\text{even}}$ , and the first equals  $(2\chi(2) - 1)B_{1,\chi}$  by the same substitution applied to  $\chi(2)fB_{1,\chi} = \sum_a a \chi(2a)$ ; so  $-\chi(2)U = (2\chi(2) - 1)B_{1,\chi}$ , i.e.  $U = (\chi(2)^{-1} - 2)B_{1,\chi}$ . Putting the three together,  $2\chi(2)T = \frac{f}{2}(B_{1,\chi} + \chi(2)U)$  gives the displayed formula. For  $\chi$  primitive and odd,  $B_{1,\chi} = -L(0, \chi) \neq 0$ .  $\square$

*Measured:* The closed formula on 1510 odd characters, over all odd  $f \leq 121$  and primitive or not, with no discrepancy; the criterion  $\chi(2) = 1$  on the 1236 of them that are primitive, also with no discrepancy. Two decoys fail as they must: the same criterion one step off the diagonal, and the same criterion for even characters.

**Corollary 8** (zeros from digits). *Let  $R_q = \{0 \leq r < q : T_q + qC(r) = 0 = T(r)\}$ . Every  $m \geq 1$  whose base- $q$  digits all lie in  $R_q$  has  $S_\chi(m) = 0$ , for both values of  $s$ .*

(a) *If  $q \equiv 1 \pmod{4}$ , then  $0, q-1 \in R_q$ , and  $S_\chi$  has at least  $2^J - 1$  zeros below  $q^J$ , hence at least  $\frac{1}{2}X^{\log 2 / \log q} - 1$  below  $X$ .*

(b) *If  $q \equiv 7 \pmod{8}$ , then  $(q-1)/2 \in R_q$ , and  $S_\chi(\frac{q^j-1}{2}) = 0$  for every  $j \geq 1$ .*

(c) *If  $q \equiv 3 \pmod{4}$ ,  $q > 3$ ,  $h(-q) = 1$  and  $s = 1$ , then  $S_\chi(z) = 0$  along  $z_0 = q$ ,  $z_{i+1} = q(z_i + 1)$ . For  $q = 3$  and  $s = 1$  the same holds with 27 in place of  $q$ :  $z_0 = 27$ ,  $z_{i+1} = 27(z_i + 1)$ .*

(d) *For every  $q$  and  $s$ : if  $S_\chi(qk) = 0$  and  $C(r) = T(r) = 0$ , then  $S_\chi(qk + r) = 0$ . When  $\chi(2) = \chi(3) = -1$  this applies with  $r = 4$ , since  $1 - 2 - 3 + 4 = 0$ .*

(e) *If  $s = -1$ , then  $S_\chi(q^2m) = q^2S_\chi(m)$  for every  $m$ : a single zero of  $S_\chi$  gives infinitely many.*

*Proof.* By induction on the number of digits, from  $S_\chi(0) = 0$  and Theorem 6. (a)  $\chi$  is even, so  $r \leftrightarrow q - r$  gives  $T_q = 0$ ; with  $C(q-1) = 0$ , both 0 and  $q-1$  lie in  $R_q$ . (b)  $\chi$  is odd and  $\chi(2) = 1$ . Theorem 7 gives  $T((q-1)/2) = 0$ , the Euler factor  $\chi(2)^{-1} - 1$  vanishing; the half sum of Jacobi

and Dirichlet in §1 gives  $C((q-1)/2) = (2 - \chi(2))h(-q) = h(-q)$ ; and  $T_q = qB_{1,\chi} = -qh(-q)$ . So  $T_q + qC = 0$ , and  $(q^j - 1)/2$  is the number whose  $j$  digits are all  $(q-1)/2$ . (c) With  $r = 0$  and  $k = z + 1$ , Theorem 6 gives  $S_\chi(q(z+1)) = q(z+1)(s\chi(z+1) - h(-q))$  when  $S_\chi(z) = 0$ ;  $S_\chi(q) = T_q + qs = q(s - h) = 0$ , and  $z_i \equiv 0 \pmod{q}$  gives  $\chi(z_i + 1) = 1$ . At  $q = 3$ ,  $T_3 = -1$ , and three steps with  $r = 0$  give  $S_\chi(27k) = 27S_\chi(k) - 27k$ ; so  $S_\chi(27) = 0$ , and  $S_\chi(z) = 0$  with  $27 \mid z$  gives  $S_\chi(27(z+1)) = 27(z+1)\chi(z+1) - 27(z+1) = 0$ . (d) Subtracting the cases  $r$  and 0 of Theorem 6,  $S_\chi(qk+r) = S_\chi(qk) + kqC(r) + T(r)$ . (e) With  $r = 0$  and  $s = -1$ ,  $S_\chi(qm) = -qS_\chi(m) + mT_q$ ; applied twice, the terms in  $T_q$  cancel.  $\square$

| $q$                 | $R_q$              | proved infinite family | example                          | zeros $\leq 10^7$ |
|---------------------|--------------------|------------------------|----------------------------------|-------------------|
| 5                   | $\{0, 4\}$         | (a), even character    | 4, 20, 24, 100, ...              | 1023              |
| 13                  | $\{0, 12\}$        | (a)                    | 12, 156, 168, ...                | 63                |
| 29                  | $\{0, 4, 24, 28\}$ | (a), four digits       | 4, 24, 28, 116, ...              | 511               |
| 7                   | $\{3\}$            | (b) and (c)            | 3, 24, 171, ...; 7, 56, 399, ... | 1680              |
| 23                  | $\{11\}$           | (b), $h = 3$           | 11, 264, 6083, 139920, ...       | 5                 |
| 11, 19, 43, 67, 163 | $\emptyset$        | (c), $h = 1$           | 11, 132, 1463, 16104, ...        | 6–93              |
| 59                  | $\emptyset$        | block, $h = 3$         | 4591439; 16708800, ...           | 1                 |

Table 2: Zeros of  $S_\chi$  for  $s = +1$ , and the infinite family that Corollary 8 proves. The last column counts all zeros, most of which the proved families do not account for: at  $q = 7$  they give about 16 of 1680, at  $q = 19$  five of 93; translates by (d) and sporadic zeros make up the rest. At  $q = 5, 13, 29$  there are no zeros outside the digit tree; the zeros do not depend on  $s$  there, nor at  $q = 37, 53, 61, 73$ , but they do at  $q = 17, 41$ . For  $q \equiv 7 \pmod{8}$  it is the Euler factor at 2 that puts  $(q-1)/2$  in  $R_q$ ; for  $q \equiv 3 \pmod{8}$  the set  $R_q$  is empty for every  $q < 10^5$  (§5), and a proved infinite family comes from class number one or from a loop of the recursion (Proposition 9 and the family at  $q = 131$ ).

Table 2 lists where the zeros come from, and Figure 3(c) draws four of the infinite families. When  $R_q$  is empty the recursion can still close a loop on a longer word of digits.

**Proposition 9** (blocks). *Let  $q \equiv 3 \pmod{4}$ ,  $q > 3$ ,  $s = 1$ ,  $h = h(-q)$  and  $D(r) = C(r) - h$ . Suppose that  $B$  has base- $q$  digits  $d_{L-1}, \dots, d_0$  with*

$$\sum_{i < L} D(d_i) = 0 \quad \text{and} \quad S_\chi(B) = hB.$$

*Then, with  $Q = q^L$ , every  $m_t = qB(1 + Q + \dots + Q^{t-1})$ ,  $t \geq 1$ , is a zero of  $S_\chi$ .*

*Proof.* With  $s = 1$  and  $T_q = -qh$ , Theorem 6 reads  $S_\chi(qk+r) = qS_\chi(k) + qkD(r) + T(r)$ . Appending the digits of  $B$  one at a time, by induction,  $S_\chi(Qk+B) = QS_\chi(k) + Qk \sum_i D(d_i) + S_\chi(B)$ , so  $S_\chi(k) = hk$  implies  $S_\chi(Qk+B) = h(Qk+B)$ ; starting from  $k = 0$ ,  $k_t = B(1 + Q + \dots + Q^{t-1})$  has  $S_\chi(k_t) = hk_t$ . Appending the digit 0,  $S_\chi(qk_t) = q(S_\chi(k_t) + k_tD(0)) = q(hk_t - hk_t) = 0$ .  $\square$

At  $q = 59$ , where  $h = 3$ , the block  $B = (1, 22, 21, 0)_{59} = 283200$  has  $\sum D = -2 + 3 + 2 - 3 = 0$  and  $S_\chi(B) = 849600 = 3B$ , so  $S_\chi$  vanishes at  $59B = 16\,708\,800$  and infinitely often after it. At  $q = 347$ , where  $h = 5$ , a block of thirteen digits does the same. Class number greater than one is therefore no obstruction to infinitely many zeros. Blocks are the simplest loops of the recursion, which carries a relation  $S_\chi(k) = uk$  to another; other loops work as well. At  $q = 131$ , where  $h = 5$ , the digit 43 has  $C(43) = 5$  and  $T(43) = 86 = 2 \cdot 43$ , so it preserves  $S_\chi(k) = 2k$ , and a final digit 27, with  $C(27) = 3$  and  $T(27) = 0$ , sends  $S_\chi(131k + 27)$  to  $262k - 262k = 0$ : every

$$27 + 43(131 + 131^2 + \dots + 131^t), \quad t \geq 0,$$

is a zero, the first being 27, 5660, 743583.

*Measured:* The family at  $q = 131$  by direct summation for  $t \leq 2$  and by Theorem 6 for  $t \leq 5$ ; Corollary 8(e) in 246 cases at  $q = 11, 19, 23$ , with  $S_\chi(m)$  by direct summation and  $S_\chi(q^2m)$  by Theorem 6.  $S_\chi(16\,708\,800) = 0$  at  $q = 59$  by direct summation; the next two members of the family, and the block at  $q = 347$ , by Theorem 6 in exact integer arithmetic.

**Corollary 10** (every order). *Let  $q$  be an odd prime,  $\chi$  an odd Dirichlet character modulo  $q$  of any order, extended completely multiplicatively by any root of unity  $\chi(q)$ , and  $S_\chi(m) = \sum_{n \leq m} n\chi(n)$ . Then  $S_\chi(\frac{q-1}{2}) = 0$  if and only if  $\chi(2) = 1$ , and in that case  $S_\chi(\frac{q^j-1}{2}) = 0$  for every  $j \geq 1$ . The characters for which this happens are the odd characters trivial on  $\langle 2 \rangle$ ; Corollary 13 counts them.*

*Proof.* Theorem 6 and its proof hold verbatim for  $\chi$ , with  $s = \chi(q)$ . By Theorem 7,  $T((q-1)/2) = \frac{q}{2}B_{1,\chi}(\chi(2)^{-1} - 1)$  with  $B_{1,\chi} \neq 0$ , which gives the equivalence at  $j = 1$ . If  $\chi(2) = 1$ , the proof of Theorem 7 gives  $C((q-1)/2) = (\chi(2)^{-1} - 2)B_{1,\chi} = -B_{1,\chi}$ , and  $T_q = qB_{1,\chi}$ , so  $(q-1)/2 \in R_q$ .  $\square$

*Measured:* Exactly in  $\mathbb{Z}[\zeta_r]$ , for every odd character with  $\chi(2) = 1$  modulo every prime  $5 \leq q \leq 130$ , of orders 2, 6, 8 and 18, with  $\chi(q) \in \{1, \zeta_r, -1\}$  and  $j \leq 3$  where  $(q^j - 1)/2 \leq 3 \cdot 10^6$ : 210 cases, all zero. None of the 817 odd characters with  $\chi(2) \neq 1$  vanishes at  $j = 1$ , and the number of odd characters with  $\chi(2) = 1$  matches  $d/t$  at all 29 primes  $5 \leq q \leq 130$ .

So membership in the generic set is proved for  $q \equiv 1 \pmod{4}$  (by evenness) and for  $q \equiv 7 \pmod{8}$  (by the prime 2 of the first lane: the zeros are manufactured by the vanishing half sum of Theorem 7), for both signs; and for  $q \equiv 3 \pmod{8}$ , where that half sum is  $qh(-q)$  for  $q > 3$  and not 0, for  $q = 3$  and the primes of class number one, 11, 19, 43, 67, 163, with  $s = 1$ , wherever the recursion closes a loop, as at  $q = 59, 131, 347$ , and, with  $s = -1$ , as soon as  $S_\chi$  has one zero.

*Measured:* Theorem 6 at every  $m \leq 10^7$  for the 19 primes  $5 \leq q \leq 73$  and for  $q = 163$ , both signs. The set  $R_q$  for all 301 primes  $5 \leq q \leq 2000$  (and, for the emptiness at  $q \equiv 3 \pmod{8}$ , for all 2408 such primes below  $10^5$ ): for the 147 with  $q \equiv 1 \pmod{4}$  it contains 0 and  $q - 1$  and is symmetric under  $r \mapsto q - 1 - r$ , of size 2 to 12; for the 78 with  $q \equiv 7 \pmod{8}$  it contains  $(q - 1)/2$ , of size 1, 3 or 5; for the 76 with  $q \equiv 3 \pmod{8}$  it is empty. The chains of (c) by direct summation, up to  $3 \cdot 10^7$ , at  $q = 7, 11, 19, 43, 67, 163$ ; with  $s = -1$  the same chain is not a chain of zeros. The Liouville function, for comparison, has the single zero  $m = 4$  up to  $10^7$ .

### 3.2 The same two sums in Conrey's sine series

For  $q \equiv 3 \pmod{4}$ ,  $q > 3$ , the sums  $C$  and  $T$  are the ingredients of the sine series that Conrey studies,

$$f_q(x) = \sum_{n \geq 1} \left(\frac{n}{q}\right) \frac{\sin 2\pi nx}{n^2}.$$

His Theorem 4 [4], stated for squarefree  $q \equiv 3 \pmod{8}$ , uses only that the character is real, odd and primitive, and holds as it stands for prime  $q \equiv 3 \pmod{4}$ ; combined with Dirichlet's class number formula, as in his Corollary 1, it reads, in the present notation and for  $q > 3$  with  $h = h(-q)$ ,

$$f_q(x) = \frac{2\pi^2}{q^{3/2}} F_q(qx), \quad F_q(y) = T(r) + y(h - C(r)), \quad r = [y],$$

so  $F_q$  is continuous, linear on each cell  $[r, r + 1]$ , and vanishes identically on that cell exactly when  $C(r) = h$  and  $T(r) = 0$ , that is, when  $r \in R_q$ . Conrey records this alternative at the end of his §6: a zero  $x$  of  $f_q$  either has  $\sum_{n \leq xq} \chi(n) = h$  and  $\sum_{n \leq xq} n\chi(n) = 0$ , in which case  $f_q$  vanishes on a whole interval, or satisfies a second, isolated condition [4]. Extended to  $q \equiv 3 \pmod{4}$ , the digit set of Corollary 8 is exactly the set of those intervals; the corollary characterises them,

and proves that they exist for every  $q \equiv 7 \pmod{8}$ : the function  $f_q$  vanishes identically on the central cell  $[\frac{q-1}{2q}, \frac{q+1}{2q}]$ , because  $(q-1)/2 \in R_q$ , and on further cells when  $R_q$  is larger — at  $q = 103$  also on  $[\frac{47}{103}, \frac{48}{103}]$  and its mirror; for  $q \equiv 3 \pmod{8}$ , the case of Conrey's positivity question, no such interval occurs for  $q < 10^5$  (Figure 3(a,b)).

*Measured:* The series itself, truncated at  $2 \cdot 10^6$  terms, at three interior points of each cell: below  $3 \cdot 10^{-12}$  on the cells 47, 51, 55 of  $q = 103$  and on the central cells of  $q = 7, 23, 31, 47$ ; between  $4 \cdot 10^{-3}$  and  $7 \cdot 10^{-2}$  in absolute value on the control cells ( $r = 46$  at  $q = 103$  and the central cells of  $q = 59, 83$ ) at every point tested other than  $x = 1/2$ , the midpoint of the central cells, where every such series vanishes, being odd and periodic.

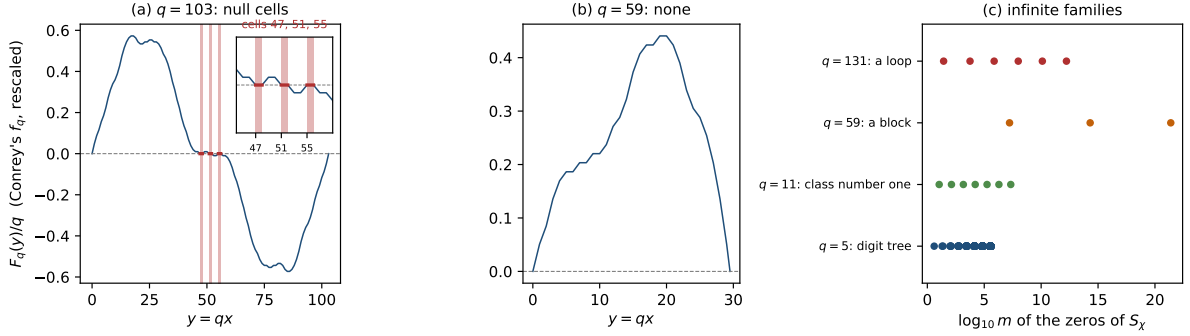


Figure 3: Zeros along one character. (a) At  $q = 103$ ,  $q \equiv 7 \pmod{8}$ , Conrey's function vanishes identically on the cells 47, 51, 55 of  $R_{103}$ , the central cell 51 and the mirror pair 47, 55: the null cells are the digits of Corollary 8. (b) At  $q = 59$ ,  $q \equiv 3 \pmod{8}$  and  $R_{59} = \emptyset$ , it stays positive on  $(0, 1/2)$  and vanishes on no cell. (c) Four infinite families of zeros of  $S_\chi$ , by the size of the zero: the digit tree of  $q = 5$ , the class-number-one chain of  $q = 11$ , the block of  $q = 59$  and the loop of  $q = 131$ ; every point is a proved zero, and all are rechecked by Theorem 6 in exact arithmetic.

**Proposition 11** (positivity empties the digit set). *Let  $q \equiv 3 \pmod{8}$ ,  $q > 3$ , and let  $\chi_{q,-1}$  be the extension with value  $-1$  at  $q$ . If  $\sum_{n \geq 1} \chi_{q,-1}(n) \sin(2\pi nx)/n^2 \geq 0$  for  $0 \leq x \leq 1/2$ , then  $R_q = \emptyset$ .*

*Proof.* By the symmetry  $C(q-1-r) = C(r)$ ,  $T(q-1-r) = q(C(r) - h) - T(r)$  it suffices to exclude  $r \leq (q-1)/2$ . Suppose such an  $r$  lies in  $R_q$  and put  $u = (rq+1)/q^2 \in [0, 1/2]$ . Splitting  $n$  by its  $q$ -adic valuation, the series at  $u$  is  $\sum_{j \geq 0} (-1)^j q^{-2j} f_q(q^j u)$ . The term  $j = 0$  vanishes,  $qu$  lying inside the cell of  $r$ ; the terms  $j \geq 2$  vanish, their arguments being integers; and the term  $j = 1$  is  $-q^{-2} f_q(1/q) = -2\pi^2 h/q^{7/2}$ , since  $F_q(1) = h$ . The series is negative at  $u$ .  $\square$

So the open question of §5 on  $R_q$  sits below a positivity statement of the kind Conrey proposes, and the digit recursion describes where his function could touch zero on a whole cell. The proposition asks for positivity of the series of  $\chi_{q,-1}$ , not of  $f_q$  itself: the term  $j = 1$  is what makes a null cell strictly negative, whereas a non-strict positivity of  $f_q$  alone would allow such cells. *Measured:* The displayed identity in floating point at  $q = 59$  and  $q = 179$ , three values of  $x$  each, to eight decimals.

## 4 Where the question came from

In [29] the ring generated by the character values of  $SU(n)$ ,  $Sp(2m)$  and  $GL(n)$  at a torsion element was shown to depend only on the segment of eigenvalue exponents, and its first layer at a prime  $p$ , for a segment stable at the prime-to- $p$  part  $q'$  of the modulus, to be the rank modulo  $p$  of a moment matrix. That matrix has a second life over  $\mathbb{Q}$ : its components at the odd

characters can vanish identically, and when they do the rank drops for every  $p$  at once. The question of exact zeros started there. Throughout this section,  $f$  plays the role of  $q'$ ,  $d = \varphi(f)/2$  is the dimension of the residual algebra, and a *cell* is a pair (segment, conductor) of a sweep.

#### 4.1 The components

Let  $f$  be odd and let the segment be  $\{-m, \dots, m\}$  with  $n = 2m + 1 \leq f$ , read modulo  $f$ . Each class is met at most once, and the moment of the class  $c$ , the sum of the exponents that fall in it, is

$$S(c) = c \quad (1 \leq c \leq m), \quad S(-c) = -c \quad (1 \leq c \leq m), \quad S(c) = 0 \text{ otherwise.}$$

The group  $G = (\mathbb{Z}/f)^\times$  acts on functions of the class by  $(u \cdot S)(c) = S(u^{-1}c)$ , and the moment matrix of [29, Thm. 10(b)] is  $\mathcal{M} = [S(u^{-1}c)]$  with a row for each  $u$ . On the diagonal  $n = f$  the segment is stable at  $f$ , the rank of  $\mathcal{M}$  modulo  $p$  is the first layer of the ring of character values, and its rank over  $\mathbb{Q}$  bounds all of them. Off the diagonal,  $n < f$ , the segment is an interval of length between 3 and  $f - 2$ , hence not stable at  $f$  [29, Lemma 13], and the order is maximal there by the support theorem [29, Thm. 16];  $\mathcal{M}$  is then a matrix of character sums, not of a ring. For a character  $\chi$  of  $G$  write

$$\hat{S}(\chi) = \sum_{c \in G} S(c) \chi(c), \quad T(m, \chi) = \sum_{c=1}^m c \chi(c).$$

**Lemma 12** (the components). *Let  $f$  be prime. Then  $\hat{S}(\chi) = (1 - \chi(-1))T(m, \chi)$ , which is  $2T(m, \chi)$  for  $\chi$  odd and 0 for  $\chi$  even, and*

$$\text{rank}_{\mathbb{Q}} \mathcal{M} = \#\{\chi \text{ odd} : T(m, \chi) \neq 0\}.$$

*In particular  $\mathcal{M}$  has rank below  $(f-1)/2$  over  $\mathbb{Q}$  exactly when some odd character has  $T(m, \chi) = 0$ , and its defect counts those characters.*

*Proof.* For  $f$  prime every nonzero class is a unit, and the column of the class 0 vanishes because  $S(0) = 0$ . The rows span the cyclic module  $\mathbb{C}[G] \cdot S$ , and rank does not change from  $\mathbb{Q}$  to  $\mathbb{C}$ . As  $G$  is abelian,  $\mathbb{C}[G] = \bigoplus_{\chi} \mathbb{C} e_{\chi}$  with  $e_{\chi} = |G|^{-1} \sum_u \overline{\chi(u)} u$ , and substituting  $v = u^{-1}c$ ,

$$(e_{\chi} S)(c) = \frac{1}{|G|} \overline{\chi(c)} \sum_{v \in G} \chi(v) S(v) = \frac{\overline{\chi(c)}}{|G|} \hat{S}(\chi),$$

so the dimension of  $\mathbb{C}[G] \cdot S$  is the number of  $\chi$  with  $\hat{S}(\chi) \neq 0$ . Splitting the sum into  $c$  and  $-c$  gives  $\hat{S}(\chi) = T(m, \chi) - \chi(-1)T(m, \chi)$ .  $\square$

Keeping one row for each pair  $\pm u$ , as [29] does, changes nothing: by the symmetry of the segment  $(-u) \cdot S = -(u \cdot S)$ . For composite  $f$  the classes that are not units form further strata, each with its own cyclic module, and the rank is the size of the union of the supports [29, §7]; the unit stratum still contributes exactly the count of the lemma.

#### 4.2 The diagonal

The moment matrix of a half-interval can be singular over  $\mathbb{Q}$ , not merely modulo  $p$ . Among all segments with  $n \leq 30$  and  $q' \leq 60$  there are 27 such cells, all with defect exactly 1, and in every one the character whose component vanishes has order 2. There is a reason for the second half of that sentence, and it is weaker than it looks: the quadratic character is the only one with rational values, hence the only one whose component can vanish *alone*; any other drags its

whole Galois orbit and the drop exceeds one. It does not say that no other character vanishes, and one row past the end of the sweep another one does. Corollary 13 below says exactly how many do, and the answer is not about the quadratic character at all: it is about the subgroup generated by 2.

By Lemma 12 the component at an odd character  $\chi$  is  $2T(m, \chi)$ , and for  $n = q'$  — the full period, which with  $f = q'$  is the diagonal  $m = (f - 1)/2$  of Figure 4 — that sum is the weighted half sum of Theorem 7, known for every odd character and used in [32, proof of Thm. 12(b)]. That formula also says what this section has to do with [29, §3], and the answer is one factor. The sawtooth of [32] has, at an odd  $\chi$ ,

$$\sum_{c=1}^{f-1} (2c - f) \chi(c) = 2f B_{1,\chi},$$

because  $\sum_c \chi(c) = 0$ . So the component of the moment matrix  $\mathcal{M}_f$  is  $B_{1,\chi}$  and the component here is  $B_{1,\chi}$  times the Euler factor at 2. On segments of length  $Nr \pm 1$  the first layer asks when  $\mathfrak{p}$  divides the first — that is the relative class number; on the diagonal it asks when  $\mathfrak{p}$  divides the second [32, Cor. 17]; and this section asks when the second is zero. Two quantities, three questions, and the whole difference between the quantities is  $\chi(2)^{-1} - 1$ .

**Corollary 13** (how many components vanish, and which). *Let  $f \geq 5$  be prime, let  $t$  be the order of 2 in  $(\mathbb{Z}/f)^\times$  and  $d = \varphi(f)/2$ . On the diagonal  $m = (f - 1)/2$  the components that vanish are exactly those at the odd characters trivial on  $\langle 2 \rangle$ , so their number is*

$$\#\{\chi \text{ odd} : T(m, \chi) = 0\} = \begin{cases} 0, & -1 \in \langle 2 \rangle, \\ d/t, & -1 \notin \langle 2 \rangle. \end{cases}$$

*Proof.* For  $f$  prime every non-trivial character is primitive, so Theorem 7 gives  $B_{1,\chi} \neq 0$  and the component vanishes exactly when  $\chi(2) = 1$ , that is when  $\chi$  factors through  $(\mathbb{Z}/f)^\times / \langle 2 \rangle$ , a group of order  $\varphi(f)/t$ . If  $-1 \in \langle 2 \rangle$  every such character is even and none counts; otherwise  $-1$  has order 2 in the quotient and exactly half of its characters are odd, which is  $\varphi(f)/2t = d/t$ .  $\square$

Primality is not a convenience. Theorem 7 buys  $B_{1,\chi} \neq 0$  from primitivity, and for composite  $f$  an imprimitive odd character can have  $B_{1,\chi} = 0$  through the Euler factors

$$B_{1,\chi} = B_{1,\chi^*} \prod_{\ell|f, \ell \nmid \text{cond}(\chi^*)} (1 - \chi^*(\ell)),$$

and then the component vanishes without  $\chi(2) = 1$ . The count is then not  $d/t$ : the smallest failure is  $f = 21$ , where the character induced from the odd character modulo 3 has  $\chi(2) = -1$  and  $T(10, \chi) = 1 - 2 + 4 - 5 - 8 + 10 = 0$ , so two components vanish where the formula predicts one. Keeping the statement for all odd  $f$  would need the factor  $B_{1,\chi}$  carried along. For the count of components that is so; for the *rank* of the moment matrix it is not, and Theorem 14 below holds for every odd  $f$ .

*Measured:* The vanishing components counted directly for every odd  $f \leq 121$ : the count is  $d/t$  at all 28 primes, and differs at 12 of the 31 composites, first at  $f = 21, 33, 39$ , never through an extra *primitive* zero — the missing hypothesis is primitivity, not the shape of the count. The cells of higher defect are values of  $d/t$ : at  $f = 31$  the order of 2 is 5 and  $d/t = 3$ , which is the defect 3 recorded in Figure 4, and the five cells there with a vanishing character of order greater than 2 —  $f = 31, 73, 89, 127, 151$  — are exactly the ones with  $d/t > 1$ .

The count of components is the wrong object for a composite  $f$ : the ring sees the whole moment matrix, with a column for every class and not only for the units, and on the whole matrix the imprimitive zeros disappear.

The statement below is known in substance: for  $f$  prime it is Jager and Lenstra's count of the dimension spanned by the conjugates of a cosecant sum [30, Thm. 3 and p. 143], and for odd  $f$  it is Girstmair's description of the annihilator of  $\text{cs}_1(1/f)$  [31, Cor. 3], where only the condition  $\chi(2) = 1$  appears. What we add is its reading as the rank of the moment matrix with every class as a column, which is the form the ring of character values sees, and a proof in that language.

**Theorem 14** (the rank on the diagonal; Jager–Lenstra, Girstmair). *Let  $f \geq 3$  be odd,  $m = (f - 1)/2$ , and let  $\mathcal{M} = [S(u^{-1}c)]$  be the moment matrix of the full period, with rows  $u \in G = (\mathbb{Z}/f)^\times$  and a column for every class  $c \in \mathbb{Z}/f$ . Then*

$$d - \text{rank}_{\mathbb{Q}} \mathcal{M} = \#\{\chi \text{ odd} : \chi(2) = 1\} = \begin{cases} 0, & -1 \in \langle 2 \rangle, \\ d/t, & -1 \notin \langle 2 \rangle, \end{cases}$$

with  $d = \varphi(f)/2$  and  $t$  the order of 2 modulo  $f$ .

*Proof.* Write  $\mathbb{Z}/f = \bigsqcup_{D|f} X_D$  with  $X_D = D \cdot (\mathbb{Z}/g)^\times$ ,  $g = f/D$ . Each stratum is  $G$ -stable, and  $G$  acts on it through its quotient  $(\mathbb{Z}/g)^\times$ . On  $X_D$  the moment is  $S(Dc') = D S_g(c')$ , where  $S_g(c')$  is the representative of  $c'$  in  $(-g/2, g/2)$ : the representative of  $Dc'$  modulo  $f$  is  $D$  times that of  $c'$  modulo  $g$ . As in Lemma 12, the dimension of  $\mathbb{C}[G] \cdot S$  is the number of characters  $\chi$  of  $G$  with  $e_\chi S \neq 0$ , and since the strata are disjoint and stable,  $e_\chi S \neq 0$  exactly when its restriction to some  $X_D$  is nonzero. Averaging over the kernel of  $G \rightarrow (\mathbb{Z}/g)^\times$  kills that restriction unless  $\chi$  factors through  $(\mathbb{Z}/g)^\times$ , that is unless the conductor  $f^*$  of  $\chi$  divides  $g$ ; and then, as in Lemma 12, it is a multiple of  $\sum_{c' \in (\mathbb{Z}/g)^\times} \chi(c') S_g(c') = (1 - \chi(-1)) T_g(\chi)$ , with  $T_g$  the half sum modulo  $g$ . For  $\chi$  even this vanishes on every stratum. For  $\chi$  odd take  $g = f^*$ : there  $\chi$  is primitive, and Theorem 7 gives  $T_{f^*}(\chi) = \frac{f^*}{2} B_{1, \chi^*}(\chi^*(2)^{-1} - 1)$  with  $B_{1, \chi^*} \neq 0$ . So  $e_\chi S \neq 0$  as soon as  $\chi(2) \neq 1$ ; and if  $\chi(2) = 1$  the same factor kills every stratum, by Theorem 7 at each  $g$ . Hence the rank is the number of odd  $\chi$  with  $\chi(2) \neq 1$ , and the defect is the number of odd  $\chi$  trivial on  $\langle 2 \rangle$ . Those are the odd characters of  $G/\langle 2 \rangle$ , a group of order  $\varphi(f)/t$ : none if  $-1 \in \langle 2 \rangle$ , and otherwise half of them,  $d/t$ .  $\square$

The imprimitive zeros of the counting are real zeros of the unit columns, and they are revived in the stratum of the conductor: at  $f = 21$  the character induced from modulus 3 vanishes on the units and not on  $X_7$ . What the ring sees is only the Euler factor at 2.

*Measured:* The rank of the full matrix, exactly over  $\mathbb{Q}$ , on all 31 odd composite  $f \leq 121$ : the defect is  $d/t$  (or 0) in 31 of 31, while the count on the unit columns differs in 12 of them; on the primes the two agree, as Lemma 12 says.

So the count of this note and the  $\delta = d/t$  that [32, Cor. 17] obtains from  $\langle 2 \rangle$  are the same number, reached from the two ends: there it measures a first layer modulo  $p$ , here it counts exact zeros over  $\mathbb{Q}$ .

**Corollary 15** (Dirichlet's family). *Let  $f \equiv 3 \pmod{4}$  be squarefree and  $\chi$  the quadratic character of conductor  $f$ , which is odd. Then*

$$\sum_{c=1}^{(f-1)/2} c \chi(c) = \begin{cases} f h(-f) \frac{2}{w(-f)}, & f \equiv 3 \pmod{8}, \\ 0, & f \equiv 7 \pmod{8}. \end{cases}$$

*Proof.* For the quadratic character  $\chi(2) = (2 | f)$ , which is 1 exactly when  $f \equiv \pm 1 \pmod{8}$ ; with  $f \equiv 3 \pmod{4}$  that is  $f \equiv 7 \pmod{8}$ , and Theorem 7 gives the vanishing. In the other case  $\chi(2) = -1$  and the value is  $-f B_{1, \chi}$ , which is  $f h(-f) 2/w(-f)$  by Dirichlet's class number formula for the imaginary quadratic field [35, §4.4].  $\square$

For prime  $f \equiv 7 \pmod{8}$  the vanishing is classical in the language of residues: the quadratic residues and non-residues below  $f/2$  have equal sums, as Berndt states [2, Cor. 13.4] and Aebi and Cairns prove elementarily [1, Thm. 1(a)].

The digit recursion of §3.1 sees this diagonal too. For the Dirichlet character itself, with  $\chi(f) = 0$ , the first term of Theorem 6 disappears, and every  $m \equiv r \pmod{f}$  with  $r \in R_f$  is a zero of  $T(m, \chi)$ : the diagonal of Figure 4 and, for even  $\chi$ , the grey family  $m = f - 1$ , repeated with period  $f$ .

*Measured:* Corollary 15 on 32 conductors  $f \equiv 3 \pmod{8}$  and 42 conductors  $f \equiv 7 \pmod{8}$ , with no discrepancy, composite  $f$  included.

So the diagonal is not a quadratic phenomenon: it is a phenomenon of the subgroup  $\langle 2 \rangle$ , and the quadratic character is merely the case where the criterion  $\chi(2) = 1$  becomes a congruence on  $f$  modulo 8. The measurement agrees. Sweeping all odd characters of prime conductor  $f < 160$  over all rows, the quadratic character vanishes in 22 cells, and 22 characters of order greater than 2 vanish, forming 6 Galois orbits at five cells,  $f = 127$  contributing two orbits — and all 22 lie on the diagonal, with  $\chi(2) = 1$ , as the theorem requires. The smallest is  $f = 31$ ,  $m = 15$ : there the quadratic component vanishes *and* so do the two components of a character of order 6, so the defect is 3. That cell is  $n = q' = 31$ , one row past the  $n \leq 30$  of the sweep.

### 4.3 Off the diagonal: the same two, at the length instead of the conductor

For  $n = 2m + 1 < q'$  the sum  $T(m, \chi)$  is short, and for a quadratic  $\chi$  it is a signed sum of the first  $m$  integers with multiplicative signs. The ring does not enter here — it is maximal off the diagonal, as recalled in §4.1 — and the rows are a question about those sums. That is enough to settle when a row can be occupied at all, and the answer is again about 2 — but at the length of the segment.

**Proposition 16** (the rows that can be occupied). *Let  $n = 2m + 1$  and let  $\varepsilon$  be any function on  $1, \dots, m$  with values  $\pm 1$ . Then*

$$\sum_{c=1}^m c \varepsilon(c) \equiv \frac{n^2 - 1}{8} \pmod{2}, \quad \text{so it can vanish only if } \left(\frac{2}{n}\right) = 1.$$

*In particular, if  $\chi$  is quadratic of conductor  $f$  with  $\gcd(f, m!) = 1$ , then  $T(m, \chi)$  does not vanish unless  $\left(\frac{2}{n}\right) = 1$  at the length  $n$  of the segment.*

Both hypotheses of that sentence are needed, and neither is cosmetic. The coprimality is what makes  $\chi$  take values  $\pm 1$  on  $1, \dots, m$ : a quadratic character vanishes at the integers it shares a factor with, and one zero in the range destroys the parity argument. At  $f = 35$ ,  $m = 6$ ,  $n = 13$  the values are  $1, -1, 1, 1, 0, -1$  and the weighted sum is  $1 - 2 + 3 + 4 + 0 - 6 = 0$  while  $\left(\frac{2}{13}\right) = -1$ ; there are 58 such pairs with  $f < 200$  and  $m \leq 40$ , and none at all once  $\gcd(f, m!) = 1$ . And  $\left(\frac{2}{n}\right)$  is the *Jacobi* symbol, which for composite  $n$  is not a criterion for 2 to be a square: at  $m = 7$ ,  $n = 15$  the pattern  $\varepsilon = (1, 1, -1, 1, -1, -1, 1)$  is multiplicative on the interval and has weighted sum 0 with  $\left(\frac{2}{15}\right) = 1$ , although the squares modulo 15 are  $0, 1, 4, 6, 9, 10$  and 2 is not among them. The correct reading is  $n \equiv \pm 1 \pmod{8}$ ; the two statements coincide only when  $n$  is prime, which is why the diagonal never showed the difference.

*Proof.* Changing one sign changes the sum by  $2c$ , so its parity is that of  $\sum_{c \leq m} c = m(m+1)/2$ , and  $m(m+1)/2 = (n^2 - 1)/8$ . The second supplementary law of quadratic reciprocity is  $\left(\frac{2}{n}\right) = (-1)^{(n^2-1)/8}$ , so the sum is even exactly when that symbol is 1.  $\square$

**Corollary 17** (the rows). *Let  $n = 2m + 1$ . The row  $m$  contains an odd quadratic character  $\chi$  of prime conductor  $f > n$  with  $T(m, \chi) = 0$  if and only if  $\left(\frac{2}{n}\right) = 1$ , and then it contains infinitely many.*

*Proof.* Necessity is Proposition 16. Conversely,  $(\frac{2}{n}) = 1$  is  $m \equiv 0, 3 \pmod{4}$ ; take  $\varepsilon$  from Theorem 1. For a prime  $f \equiv 3 \pmod{4}$ ,  $f > n$ , the value  $(2 | f)$  is fixed by  $f \pmod{8}$ , and for an odd prime  $p \leq m$  reciprocity gives  $(p | f) = (-1)^{(p-1)/2}(f | p)$ , fixed by  $f \pmod{p}$  and taking both signs. By the Chinese remainder theorem and Dirichlet's theorem there are infinitely many primes  $f \equiv 3 \pmod{4}$  with  $(p | f) = \varepsilon(p)$  for every prime  $p \leq m$ ; for them  $\chi = (\cdot | f)$  agrees with  $\varepsilon$  on  $1, \dots, m$ , and  $T(m, \chi) = S_\varepsilon(m) = 0$ .  $\square$

The diagonal is the case  $n = q'$ : there  $(\frac{2}{n}) = \chi(2)$  for the quadratic character and the proposition becomes Corollary 15. So the guess that  $\langle 2 \rangle$  governs off the diagonal too is not wrong; it is only asked at the wrong modulus. It governs at  $n$ , not at  $f$ , and the two coincide only on the diagonal. What the row then contains is a finite question: the quadratic components that vanish in row  $m$  are those whose sign pattern on the primes  $\leq m$  solves  $\sum_{c \leq m} c \varepsilon(c) = 0$  multiplicatively, and each solution is a set of Legendre conditions, that is a congruence on  $f$ .

*Measured:* The converse of Proposition 16, proved for every row in Corollary 17, first checked in the 20 rows  $2 \leq m \leq 40$  with  $(\frac{2}{n}) = 1$ : in each a multiplicative solution exists, and the row is occupied; with the 19 rows the proposition empties, the pattern is right in 39 of 39. A decoy with  $(\frac{3}{n})$  in place of  $(\frac{2}{n})$  agrees on only 19 of the 39, which is chance. The characterisation of *which*  $f$  occur was checked against the odd quadratic characters of squarefree conductor  $f \equiv 3 \pmod{4}$  with  $4m < f < 400$  and  $\gcd(f, m!) = 1$  — the hypothesis of the proposition — in 787 cells  $(m, f)$  with  $m \leq 20$ , with no discrepancy; the row  $m = 3$  comes out as the single solution  $1 + 2 - 3 = 0$ , that is  $(2 | f) = 1$  and  $(3 | f) = -1$ , which is what Figure 4 shows.

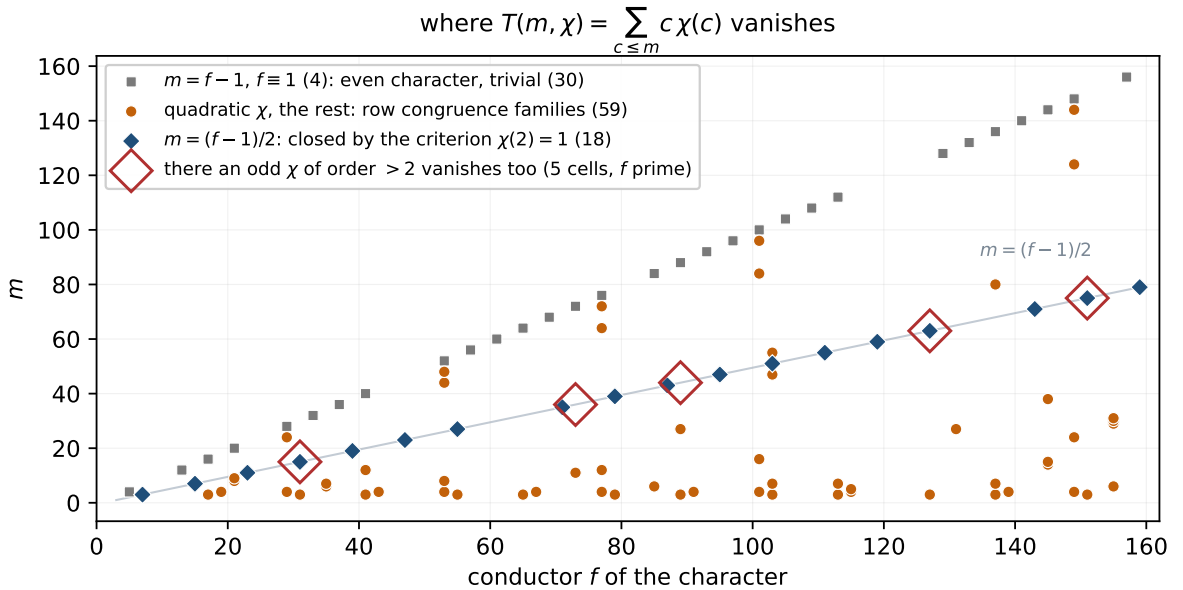


Figure 4: Where  $T(m, \chi)$  vanishes. The blue diamonds are the diagonal  $m = (f - 1)/2$ , closed by Theorem 7: there the vanishing is exactly  $\chi(2) = 1$ , which for the quadratic character is  $f \equiv 7 \pmod{8}$ . The grey squares are the trivial family  $m = f - 1$  with  $f \equiv 1 \pmod{4}$ , where the character is even and the pairing  $c \leftrightarrow f - c$  kills the sum by itself; they lie outside the range  $n \leq f$  of the components and are drawn for comparison. The orange dots are the quadratic zeros that remain: they fall into horizontal rows — the row  $m = 3$ , for instance, is the set of  $f$  with  $(2 | f) = 1$  and  $(3 | f) = -1$  — so each small  $m$  imposes its own congruence on  $f$ , and Corollary 17 says that every admissible row is occupied. The five large red diamonds are the cells where an odd character of order greater than 2 also vanishes; the leftmost is  $f = 31$ ,  $m = 15$ , where the defect is 3 and not 1.

#### 4.4 Off the diagonal, characters of higher order

A character of order  $r > 2$  drags its Galois orbit, and one might expect it never to vanish off the diagonal. It does, and for two different reasons.

**Proposition 18** (higher order off the diagonal). *Let  $f$  be prime,  $\chi$  an odd character modulo  $f$  of order  $r > 2$ , and  $1 \leq m < (f - 1)/2$ .*

- (a) *If  $\chi$  takes only the values  $\pm 1$  on  $1, \dots, m$ , then  $T(m, \chi) = 0$  exactly when the sign pattern  $\varepsilon = \chi|_{[1, m]}$  solves the row  $m$ , that is  $\sum_{c \leq m} c \varepsilon(c) = 0$ .*
- (b) *If  $r = 4$ , then  $\chi^2$  is the Legendre symbol,  $\chi$  takes values  $\pm 1$  on the quadratic residues and  $\pm i$  on the non-residues, and  $T(m, \chi) = 0$  exactly when the two signed sums, over the residues  $c \leq m$  and over the non-residues  $c \leq m$ , vanish separately. In particular both  $\sum_{c \leq m, (c|f)=1} c$  and  $\sum_{c \leq m, (c|f)=-1} c$  must be even.*

*Proof.* (a) is the definition. (b)  $T(m, \chi) = R + iI$  with  $R, I \in \mathbb{Z}$  the two signed sums, and  $1, i$  are linearly independent over  $\mathbb{Q}$ ; changing a sign changes a sum by an even amount, which gives the parity condition.  $\square$

So a zero with real prefix is a solution of a row, realised by a character of higher order: the pattern  $\varepsilon(2) = \varepsilon(3) = -1$  of the row  $m = 4$ ,  $1 - 2 - 3 + 4 = 0$ , is realised by a character of order 6 at  $f = 307, 499$  and  $643$ . A zero with non-real prefix is a new phenomenon, two rows solved at once on the two classes of the Legendre symbol: at  $f = 421$ ,  $m = 20$ , with the primitive root  $g = 2$  and  $\chi(2) = i$ , the residues  $\{1, 3, 4, 5, 7, 9, 11, 12, 15, 16, 17, 20\}$  and the non-residues  $\{2, 6, 8, 10, 13, 14, 18, 19\}$  both carry signed sums equal to 0 (their plain sums are 120 and 90).

*Measured:* All odd characters of order  $r > 2$  of prime conductor  $f \leq 1200$ , one representative per Galois orbit, tested exactly by the divisibility  $\Phi_r(X) \mid \sum_{n \leq m} n X^{\log_g n \bmod r}$ : seven zeros off the diagonal with real prefix,  $(f, m, r) = (307, 4, 6), (439, 3, 6), (499, 4, 6), (643, 4, 6), (727, 3, 6), (919, 3, 6), (937, 3, 8)$ , and one with non-real prefix,  $(421, 20, 4)$ . None below  $f = 307$ .

## 5 What is open

▷ **Class number greater than one.** For  $q \equiv 3 \pmod{8}$  the set  $R_q$  is empty for every  $q < 10^5$ , and we have no proof that it always is; by Proposition 11 it holds for each  $q$  at which the series of  $\chi_{q,-1}$  is non-negative on  $[0, 1/2]$ . When moreover  $h(-q) > 1$  and  $s = 1$ , no mechanism of Corollary 8 applies, and the zeros are few: up to  $10^7$ ,  $q = 59$  has one,  $q = 179, 227, 347$  none, and no  $q < 400$  more than eight; at  $q = 179$  and  $227$  the recursion, run as an automaton on the digits, finds none with at most 40 digits. Yet at  $q = 59, 131, 347$  a loop of the recursion gives infinitely many. Does every  $\chi_{q,1}$  with  $q \equiv 3 \pmod{8}$  close a loop — and so have infinitely many zeros — or are there  $q$ , perhaps 179 or 227, with none at all? For  $s = -1$  one zero is enough, by Corollary 8(e); what is open there is whether every  $\chi_{q,-1}$  has one.

▷ **Values in  $\mu_4$ .** Theorem 1 for functions with values in  $\{\pm 1, \pm i\}$  is immediate. With  $\psi(2) = i$  imposed, so that the function  $\psi$  is not real on the first prime — as for the character of order 4 of conductor 421 in Proposition 18 — it is not: among the lengths  $m \equiv 0, 3 \pmod{4}$  there is a solution at  $m = 12$ , none at 15 and 16, and one at every  $19 \leq m \leq 3000$ . The primes of  $(m/2, m]$  enter once, now with weights  $\pm q$  or  $\pm iq$ , so splitting them between the real and the imaginary part reduces each row to two copies of (1). Is there a solution with  $\psi(2) = i$  at every  $m \geq 19$  with  $m \equiv 0, 3 \pmod{4}$ ?

▷ **How many solutions.** The number  $a(m)$  of sign patterns on the primes  $\leq m$  with  $S_\varepsilon(m) = 0$  begins  $1, 1, 1, 1, 2, 2, 2, 4, 1, 9, 5, 14, \dots$  on the admissible lengths. It is not in the OEIS, where a related unweighted sequence is [11]. Its growth is described by a mixture, not by a single Gaussian (Figure 5), and we cannot prove it.

*Measured:* The count  $a(m)$ , exactly for  $m \leq 120$ . Choosing the signs of the primes at random, fixing those up to  $\sqrt{m}$  leaves a sum of independent terms  $\pm A_p$ ,  $A_p = p S_\varepsilon(\lfloor m/p \rfloor)$ ,  $p > \sqrt{m}$ , around a value  $R$  with variance  $V = \sum A_p^2$ , both depending on the small signs; the conditional Gaussian prediction  $2^{\pi(m)} \mathbb{E}[2(2\pi V)^{-1/2} e^{-R^2/2V}]$ , with the expectation computed exactly, matches  $a(m)$  within 8% for every admissible  $67 \leq m \leq 120$  and within 3% from  $m = 80$ . The plain Gaussian with the global mean and variance undercounts by a nearly constant factor, between 1.30 and 1.54 in that range: the density of a mixture of scales at its centre exceeds that of the Gaussian with the averaged variance.

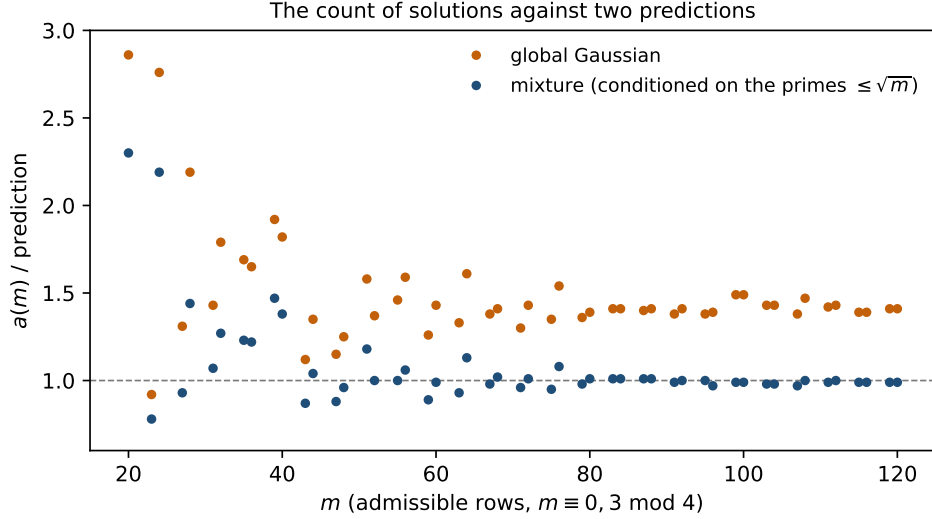


Figure 5: The number of solutions  $a(m)$  against two predictions. The global Gaussian, with the mean and variance of all sign choices, falls short by a nearly constant factor; the mixture, conditioned on the signs of the primes up to  $\sqrt{m}$ , settles at 1.

**Remark 19** (the order 2 is critical). For  $f(p)$  independent and uniform in  $\mu_r$ , extended completely multiplicatively,  $\mathbb{E} f(n) \overline{f(k)} = 1$  exactly when  $n/k$  is an  $r$ -th power of a rational, and writing  $n = da^r$ ,  $k = db^r$  with  $\gcd(a, b) = 1$ ,

$$\mathbb{E} \left| \sum_{n \leq m} n f(n) \right|^2 = \sum_{\gcd(a,b)=1} a^r b^r \sum_{d \leq m / \max(a,b)^r} d^2,$$

which is  $\sim C_r m^3$  for  $r \geq 3$  but of order  $m^3 \log m$  for  $r = 2$ . The logarithm in Lemma 2 is a feature of real signs. *Measured:* The identity against the exact average over all  $r^{\pi(m)}$  assignments,  $r = 2, 3, 4$ ,  $m \leq 20$ ;  $C_4 = 0.3402817\dots$  by the series and by the limit at  $m = 10^{10}$ .

## MSC 2020

Primary 11A25; Secondary 11K65, 11L40, 11P70, 11B85, 11R29.

## Declaration of competing interest

The author declares no competing interest and no funding.

## Data availability

No external data were used. Every number labelled *measured*, and every figure, is generated by scripts that travel with this note together with their archived output: among them the generator and checker of the 104 block certificates of Theorem 1, which runs in integer arithmetic in seconds, the inequalities of its analytic tail, and the Lean 4 file that formalizes Theorem 6, the propagation of zeros of Corollary 8 and Lemma 3(b). They are released into the public domain under CC0.

## Declaration on generative AI

This note was written with Claude (Anthropic) as an assistant, under the author's direction. Nothing in it rests on the assistant's word: the finite range of Theorem 1 rests on integer certificates that an archived script regenerates and checks, the constants of its analytic tail are recomputed by another, the digit recursion and the propagation of its zeros are formalized in Lean, and every statement marked *measured* is produced by a script archived with the note. The author is responsible for the content.

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